



CANDIDATE
NAME

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NUMBER

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CANDIDATE
NUMBER

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9702/36

October/November 2021

2 hours

You must answer on the question paper.

You will need: The materials and apparatus listed in the confidential instructions

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You will be allowed to work with the apparatus for a maximum of 1 hour for each question.
- You should record all your observations in the spaces provided in the question paper as soon as these observations are made.
- You may use a calculator.
- You should show all your working and use appropriate units.

- The total mark for this paper is 40.
- The number of marks for each question or part question is shown in brackets [].

For Examiner's Use	
1	
2	
Total	

This document has **16** pages. Any blank pages are indicated.

You may not need to use all of the materials provided.

1 In this experiment, you will investigate the oscillations of a wooden strip.

(a) (i) • Assemble the apparatus as shown in Fig. 1.1 with the nail held securely in the boss.

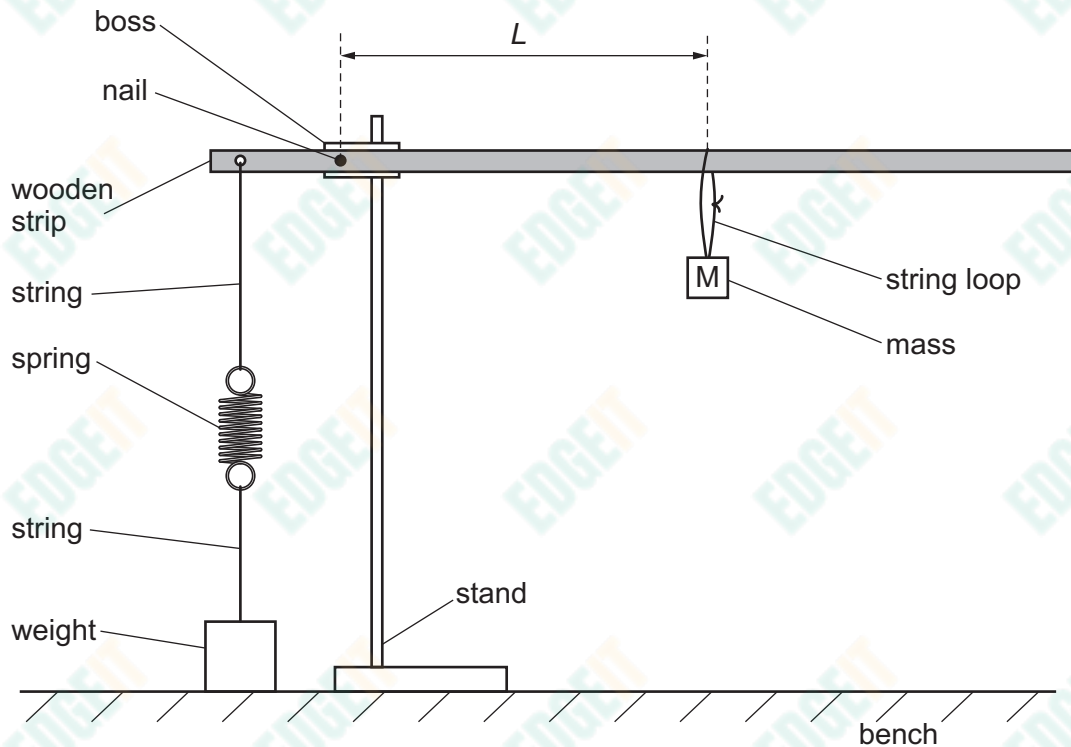


Fig. 1.1

- Hang the mass labelled M midway between the nail and the end of the strip.
- Adjust the height of the boss so that the strip is parallel to the bench.
- Move the weight so that the spring is vertical.
- L is the distance between the nail and the string loop attached to M , as shown in Fig. 1.1.

Measure and record L .

$L = \dots\dots\dots$ [1]

- (ii) • Pull the free end of the strip down by approximately 2 cm. Release the strip so that it oscillates.
- Take measurements to determine the period T of the oscillations.

$T = \dots\dots\dots$ s [2]

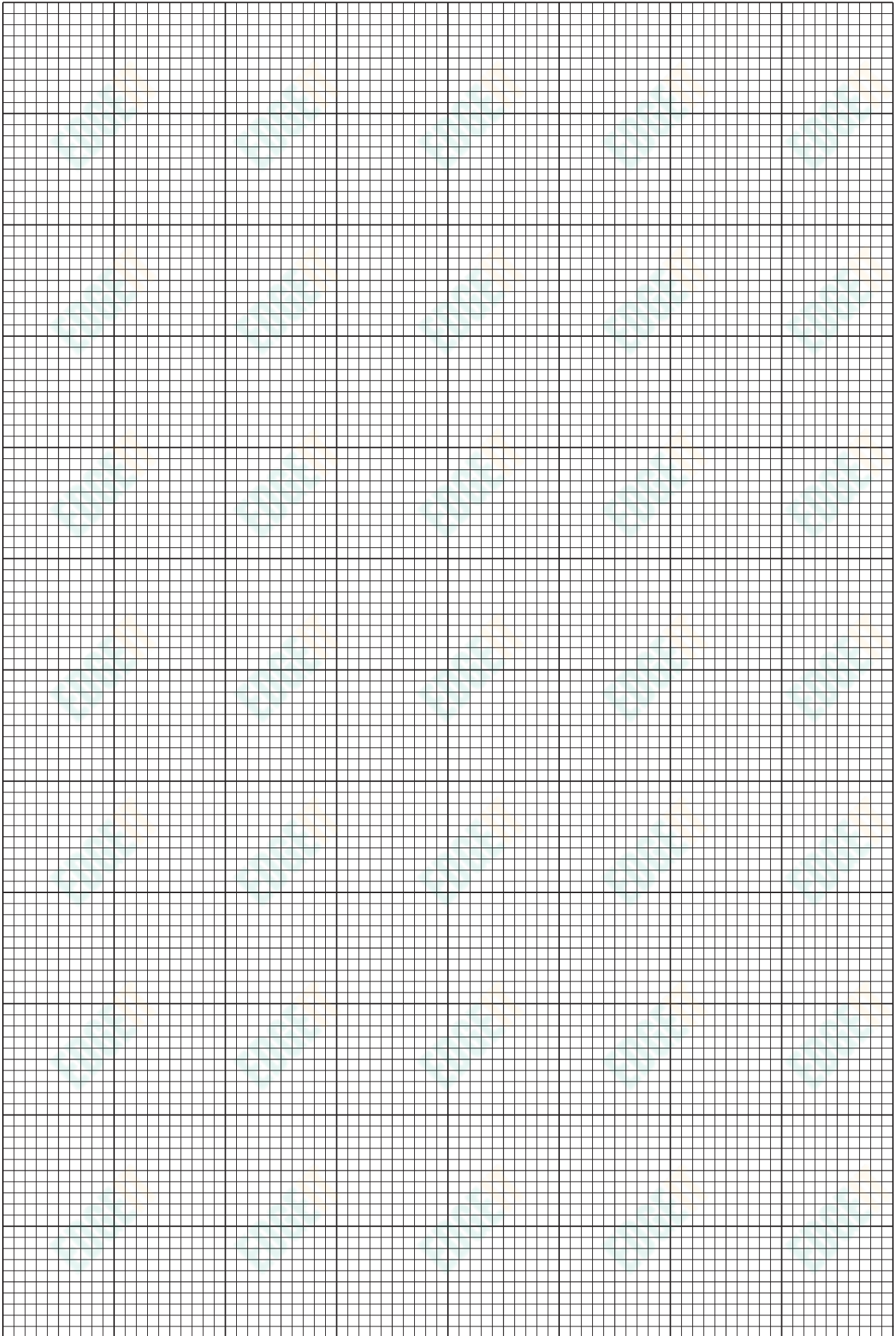
- (b)
- Move M along the strip and adjust the apparatus so that the strip is parallel to the bench.
 - Measure L and determine T .
 - Repeat until you have six sets of values of L and T .
Record your results in a table. Include values of L^2 and T^2 in your table.

- (c)
- (i) Plot a graph of T^2 on the y-axis against L^2 on the x-axis. [9]
- (ii) Draw the straight line of best fit. [3]
- (iii) Determine the gradient and y-intercept of this line. [1]

gradient =

y-intercept =

[2]



- (d) It is suggested that the quantities T and L are related by the equation

$$T^2 = aL^2 + b$$

where a and b are constants.

Use your answers in (c)(iii) to determine the values of a and b .
Give appropriate units.

$a =$

$b =$

[2]

[Total: 20]

You may not need to use all of the materials provided.

2 In this experiment, you will investigate the amount of air needed to lift an underwater load.

(a) You are provided with a syringe attached to a long tube containing a wire.

- Bend the end of the tube as shown in Fig. 2.1.

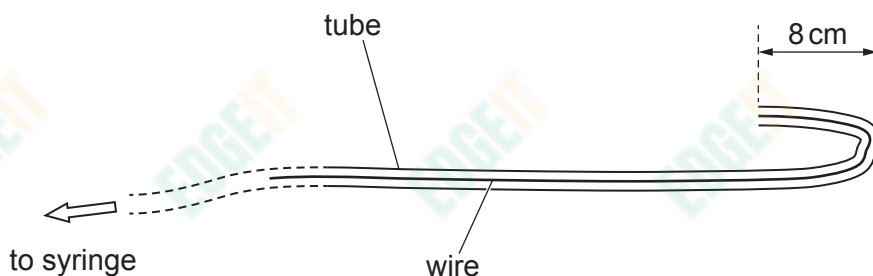


Fig. 2.1

- Pull the plunger of the syringe to the 50 cm^3 mark ($1\text{ cm}^3 = 1\text{ ml}$).
- Bend the tube and hook it over the container so that the end is approximately 11 cm above the bottom of the container, as shown in Fig. 2.2.

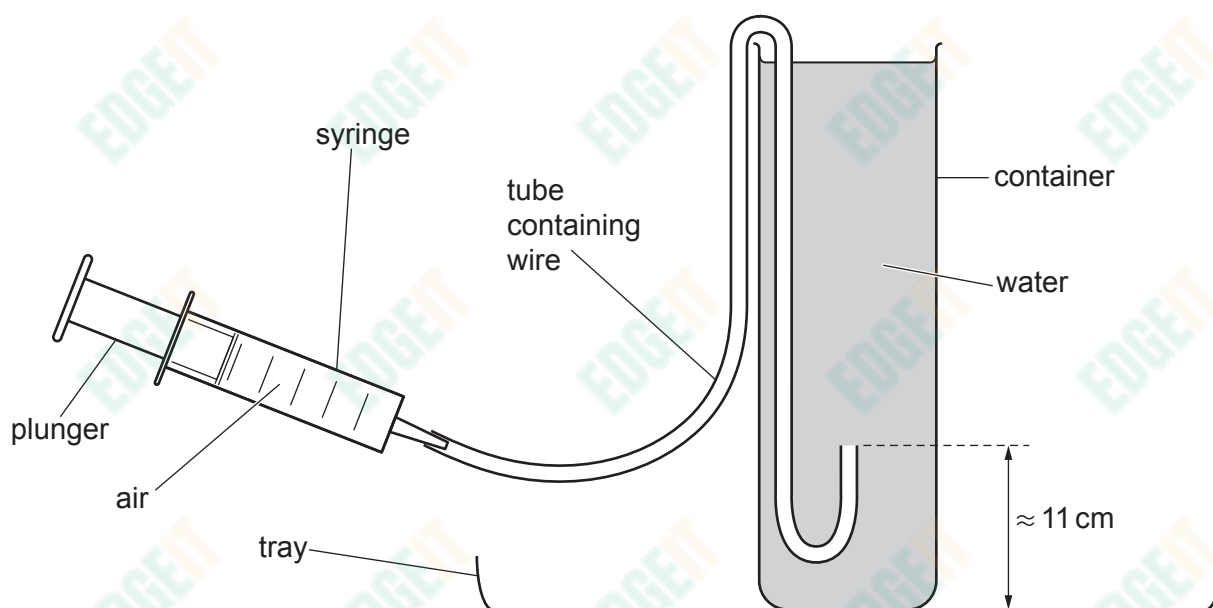


Fig. 2.2

- (i) You have been provided with a set of metal rings.

Take measurements to determine the average thickness t of the rings.
Show your working.

$t =$ cm [1]

- (ii) Measure and record the inner diameter d_1 and the outer diameter d_2 of one of the metal rings, as shown in Fig. 2.3.

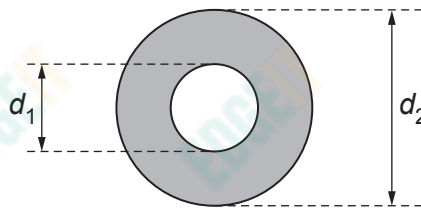


Fig. 2.3

$d_1 =$ cm

$d_2 =$ cm [1]

- (iii) Calculate the volume V_R of a metal ring using

$$V_R = \frac{\pi t(d_2^2 - d_1^2)}{4}.$$

$V_R =$ cm³ [1]

- (iv) Justify the number of significant figures that you have given for your value of V_R .

.....

 [1]

(b) You have been provided with a paper clip and a plastic cup with string attached.

- (i) • Bend the paper clip into a hook shape as shown in Fig. 2.4. Add 8 metal rings to the paper clip and hook it onto the string loop, as shown in Fig. 2.4.

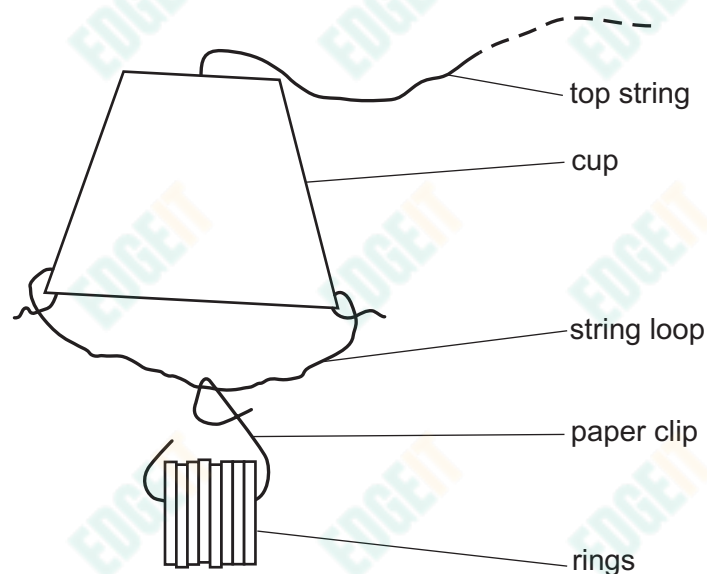


Fig. 2.4

- Lower the cup into the water. Ensure that the cup is **completely** filled with water.
- Use the top string to position the cup over the end of the tube, as shown in Fig. 2.5.

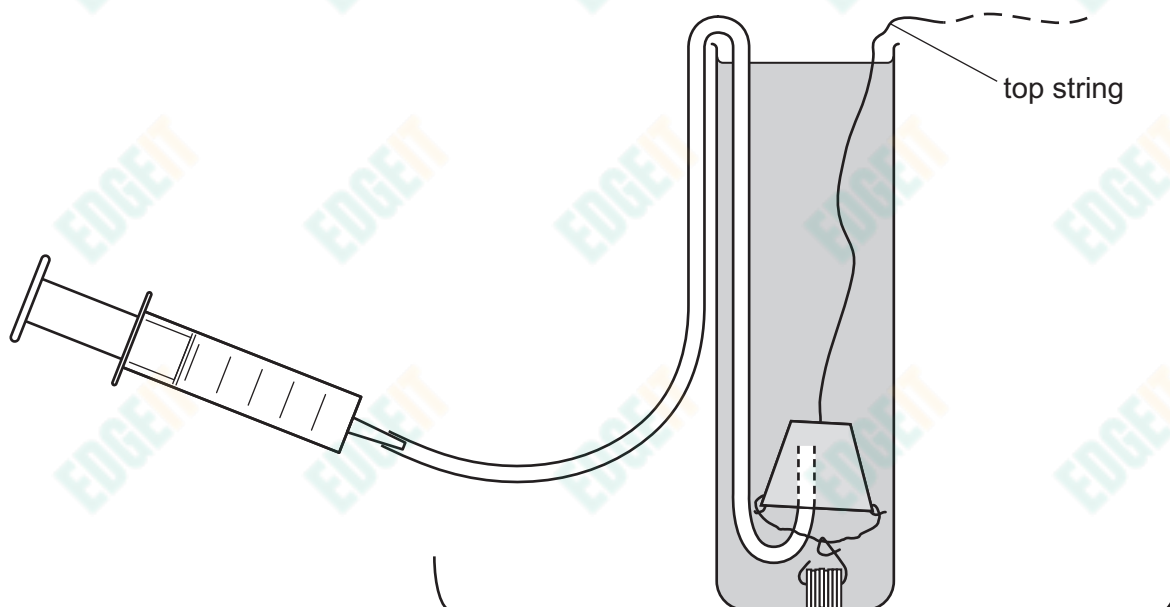


Fig. 2.5

- Record the initial reading x_1 from the syringe scale.

$x_1 = \dots\dots\dots \text{cm}^3$ [1]

- (ii)
- Slowly press the plunger of the syringe so that air enters the cup. Continue until the cup starts to lift the rings towards the surface of the water.
 - Record the final reading x_2 from the syringe scale.

$$x_2 = \dots\dots\dots \text{cm}^3$$

- Calculate the volume of air V_A in the cup using

$$V_A = x_1 - x_2.$$

$$V_A = \dots\dots\dots \text{cm}^3$$

[2]

- (iii) Estimate the percentage uncertainty in your value of V_A . Show your working.

$$\text{percentage uncertainty} = \dots\dots\dots [1]$$

- (c)
- Remove the cup and the tube from the container of water.
 - Pull the plunger of the syringe to the 50 cm^3 mark to draw air into the syringe.
 - Replace the tube in the container of water.
 - Repeat **(b)(i)** and **(b)(ii)** using 12 metal rings.

$$x_1 = \dots\dots\dots \text{ cm}^3$$

$$x_2 = \dots\dots\dots \text{ cm}^3$$

$$V_A = \dots\dots\dots \text{ cm}^3$$

[2]

(d) (i) The mass M of one metal ring is given on the card.

- Write down the value of M .

$M = \dots\dots\dots$ g

- The number of rings attached to the paper clip is n .
It is suggested that the relationship between n , M , V_R and V_A is

$$nM = k(nV_R + V_A)$$

where k is a constant.

Using your data, calculate two values of k .

first value of $k = \dots\dots\dots$

second value of $k = \dots\dots\dots$ [1]

(ii) Explain whether your results support the suggested relationship.

.....

 [1]

(e) (i) Describe four sources of uncertainty or limitations of the procedure for this experiment.

1.
.....
 2.
.....
 3.
.....
 4.
.....
- [4]

(ii) Describe four improvements that could be made to this experiment. You may suggest the use of other apparatus or different procedures.

1.
.....
 2.
.....
 3.
.....
 4.
.....
- [4]

[Total: 20]

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